Sequential Response Surface Methodology and Central Composite Design for Multi-Objective Hyperparameter Optimization in Gradient Boosted Trees under Stochastic Nuisance Blocking

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Abstract

Hyperparameter optimization in gradient boosted decision trees is commonly conducted using stochastic black-box search methods, which offer limited parametric interpretability and often disregard data-partitioning nuisance variation. We implement a classical Design of Experiments (DOE) framework based on Douglas C. Montgomery's methodology to optimize an XGBoost regressor across four hyperparameters under stochastic nuisance blocking. A Randomized Complete Block Design across five deterministic seed blocks isolates nuisance variance ($\sigma_{\text{block}} \approx 0.0077$ RMSE, ICC = 0.4069). A 140-run Face-Centered Central Composite Design detects significant response curvature ($F = 105432.31$) and structural lack of fit ($F = 62.21$), confirming that second-order polynomial surrogates provide effective local guidance maps but exhibit structural optimism at intermediate tree depths. Canonical spectral analysis reveals a stationary ridge system, and multi-objective Derringer-Suich desirability optimization identifies an operational compromise coordinate ($\mathbf{x}^*_{\text{MO}}$, depth 4) balancing validation error against inference latency. We evaluate the framework through a prospective benchmark campaign comprising 17,077 model fits and 1,640,810 timed inferences across 20 independent search replicates per optimizer. Winning configurations (122 selection records, 95 distinct configurations) were frozen in `finalized_selections.json` before evaluation across 20 retraining seeds on the external holdout test set ($N = 4{,}128$), enforcing programmatic pipeline isolation during the revised search while recognizing that the same dataset was previously evaluated in historical baseline experiments. Single-objective TPE achieves the lowest observed test RMSE (0.46691) at high latency (188.93 μs), whereas DOE Single-Objective yields deterministic selection (0.46859) with 20.2% lower latency. In multi-objective regimes, Repeated DOE and Multi-Objective TPE show no statistically significant difference in test RMSE (0.48997 vs. 0.49255, $p = 0.38$) at comparable inference latencies ($\approx 122\,\mu$s, 100% constraint feasibility). While MO-TPE achieves higher candidate hypervolume on development split 42 (18.3834 vs. 16.8585), holdout evaluation reveals a 12-point non-dominated set spanning both DOE and TPE configurations. These results establish DOE as a statistically grounded, interpretable paradigm for tabular hyperparameter optimization.

Keywords: Design of Experiments (DOE), Response Surface Methodology (RSM), Central Composite Design (CCD), Gradient Boosted Trees, XGBoost, Multi-Objective Optimization, Randomized Complete Block Design (RCBD), Canonical Analysis, Model Adequacy.

1 Introduction and Theoretical Framework

Gradient Boosted Decision Trees (GBDTs), such as XGBoost [2], achieve competitive predictive accuracy on tabular data. However, tuning GBDT hyperparameters involves non-linear

interactions, non-convex loss surfaces, and operational trade-offs between predictive accuracy and single-sample inference latency.

Standard practice relies on heuristic search algorithms, including uniform random search [3] and sequential Bayesian optimization methods such as Tree-structured Parzen Estimators (TPE) [4, 5] or Gaussian Process regression. In practical workflows, these black-box optimizers present three structural trade-offs relative to parametric response surface methods:

(i) **Absence of Parametric Attribution:** Black-box optimizers treat the objective function as an opaque evaluation oracle. They cannot quantify the percentage of variance explained by main effects versus two-factor interactions, nor do they provide an analytical surrogate for sensitivity analysis.

(ii) **Confounding of Stochastic Nuisance:** Variation arising from random data partitioning and stochastic subsampling is confounded with hyperparameter sensitivity unless structured blocking is introduced.

(iii) **Lack of Curvature and Adequacy Tests:** Heuristic search strategies lack hypothesis tests to determine whether an operational region is planar, quadratic, or deficient in fit.

To address these limitations within an interpretable framework, this study implements a sequential Design of Experiments (DOE) pipeline based on Chapters 5, 9, 10, and 14 of Douglas C. Montgomery's *Design and Analysis of Experiments* [1], following response surface methodology developed by Box and Wilson [7] and extended by Myers et al. [8]. By combining randomized complete block screening, curvature testing, Central Composite Design (CCD) augmentation, spectral canonical analysis, and Derringer-Suich desirability functions [6], we evaluate the extent to which classical response surface methodology can optimize GBDTs.

All metrics, sums of squares, test statistics, and diagnostics reported in this paper are derived from model fits on the California Housing benchmark ($N = 20,640$). An external holdout test set of 4,128 observations (20%) was set aside from the 16,512-observation development pool, which each seed block partitioned into 12,384 training (75%) and 4,128 validation (25%) samples. Whereas the historical $v1.0.0$ experiments recorded test results during development, the revised pipeline prevented access to those test labels during all surrogate fitting, curvature testing, lack-of-fit analysis, and desirability optimization until winning configurations were frozen in `finalized_selections.json`. These safeguards enforce programmatic holdout isolation during the revised search, although they cannot retroactively undo earlier exposure of the same dataset.

2 Experimental Architecture and Mathematical Formulation

2.1 Operational Factors and Standard Coding

We define four hyperparameter factors (three continuous, one integer depth) controlling structural complexity and regularization for an XGBoost regressor with $n_{\text{estimators}} = 100$:

- **Factor A (x_1):** Learning rate ($\eta \in [0.01, 0.30]$), coded on a logarithmic scale: $\xi_1 = \ln(\eta) \in [\ln(0.01), \ln(0.30)] = [-4.6052, -1.2040]$.
- **Factor B (x_2):** Maximum tree depth (depth $\in [3, 9]$), controlling structural model complexity.
- **Factor C (x_3):** Subsample fraction (subsample $\in [0.50, 1.00]$), introducing stochastic row subsampling.
- **Factor D (x_4):** L_2 leaf regularization ($\lambda \in [0.10, 10.00]$), coded on a logarithmic scale: $\xi_4 = \ln(\lambda) \in [\ln(0.10), \ln(10.00)] = [-2.3026, 2.3026]$.

Each natural variable ξ_i is mapped to dimensionless coded space $x_i \in [-1, +1]$ via:

$$x_i = \frac{\xi_i - \frac{\xi_{i,\max}+\xi_{i,\min}}{2}}{\frac{\xi_{i,\max}-\xi_{i,\min}}{2}}, \quad i \in \{1, 2, 3, 4\} \tag{1}$$

The operational domain forms a 4-dimensional hypercube $\mathcal{D} = [-1, +1]^4$. Table 1 summarizes factor coding.

Factor	**Natural Variable**	**Transform**	**Low −1**	**Center 0**	**High +1**
Factor A (x_1)	Learning Rate (η)	$\ln(\eta)$	0.0100	0.0548	0.3000
Factor B (x_2)	Max Tree Depth	Linear	3	6	9
Factor C (x_3)	Subsample Fraction	Linear	0.5000	0.7500	1.0000
Factor D (x_4)	L_2 Reg (λ)	$\ln(\lambda)$	0.1000	1.0000	10.0000

Table 1: Hyperparameter Factors, Operational Domains, and Design Levels

2.2 Randomized Complete Block Design (RCBD) for Stochastic Nuisance

Stochastic variation in data partitioning and subsampling introduces non-trivial variance across evaluation trials. We structure the experimental protocol as a Randomized Complete Block Design (RCBD) across $b = 5$ distinct seed blocks: $\mathcal{S} = \{42, 101, 202, 303, 404\}$. Table 2 summarizes the assignment of split seeds and model seeds across phases.

Phase / Design Point	**Point IDs**	**Block**	**Split Seed**	**Model Seed**	**Reps**	**Isolated Variance**
Phase 1: Corner Runs	$1 \ldots 16$	$b \in \{1..5\}$	S_b	S_b	1	Factorial + Block
Phase 1: Center Points	17	$b \in \{1..5\}$	S_b	$S_b + 1000r$	4	Subsampling Error (15 df)
Phase 2: Axial Runs	$18 \ldots 25$	$b \in \{1..5\}$	S_b	S_b	1	Quadratic + Block

Table 2: Structure of Blocks, Split Seeds, and Model Seeds Across Experimental Phases

Specifically, each block b defines a distinct 75/25 train/validation partition of the 16,512 development observations via split seed S_b. For corner runs ($1 \ldots 16$) and axial runs ($18 \ldots 25$), the internal XGBoost subsampling seed is also set to S_b, meaning that the block factor captures the joint nuisance impact of data-split selection and stochastic subsampling realization. For the four center point replicates within block b, the train/validation split is held fixed at S_b while the model subsampling seed is varied via $S_b + r \times 1000$ ($r \in \{0, 1, 2, 3\}$), isolating within-block stochastic subsampling pure error (15 degrees of freedom) from block-to-block variation (4 degrees of freedom).

The statistical model for response y_{uj} at design point u in block j is:

$$y_{uj} = \mu + \tau_u + \beta_j + \epsilon_{uj}, \quad j \in \{1, \ldots, 5\} \tag{2}$$

where τ_u represents the treatment effect of design coordinate $\mathbf{x}_u$, β_j represents the block effect, and ϵ_{uj} is experimental error.

To quantify the magnitude of variance absorbed by blocking, we compute the Intraclass Correlation Coefficient (ICC) under both ANOVA and REML formulations:

$$\text{ICC} = \frac{\sigma^2_{\text{block}}}{\sigma^2_{\text{block}} + \sigma^2_{\epsilon}} \tag{3}$$

For validation RMSE, the ANOVA estimator yields ICC = 0.4069 ($F = 20.21, p < 0.0001$), which matches the restricted maximum likelihood estimate ($\text{ICC}_{\text{REML}} = 0.4069$). Importantly, this 40.7% represents a share of *unexplained residual variance* rather than total experimental variation (where primary factors account for $> 90\%$ of total sum of squares). In absolute terms, the estimated block standard deviation is $\sigma_{\text{block}} = \sqrt{\sigma^2_{\text{block}}} \approx 0.0077$ RMSE. This nuisance variation is comparable in magnitude to the performance differences separating competitive hyperparameter configurations (often 0.005–0.015 RMSE), which makes blocking necessary when comparing close configurations.

While the block effect is non-significant in Phase 1 ($F = 0.33, p = 0.8541$), it is significant in the Phase 2 second-order model ($F = 20.21, p < 10^{-11}$). This difference reflects variance shielding: in Phase 1, unmodeled quadratic curvature pooled into the residual denominator ($\text{MS}_E = 0.003485$), inflating error variance and attenuating the block F-ratio. Once second-order terms accounted for curvature in Phase 2 (MS_E reduced to 0.000086), seed partition variance became detectable. For inference latency, the block effect is negligible ($\text{ICC} = 0.0000, F = 0.75, p = 0.5591$), as execution speed is governed by hardware architecture and tree topology rather than data splits.

2.3 Dual Responses and Latency Measurement Protocol

For every trial, two responses are measured:

1. **Response Y_1 (Validation RMSE):** Root mean squared error evaluated on the 25% validation partition ($N_{\text{val}} = 4,128$) of the development pool. Unlike the historical $v1.0.0$ pipeline, which recorded test metrics during development, the revised pipeline prevented access to the holdout 20% test partition ($N_{\text{test}} = 4,128$) until candidate configurations were frozen for final generalization testing.

2. **Response Y_2 (Single-Sample Inference Latency):** Execution time in microseconds per single sample (μs/sample) on CPU. We emphasize that this represents single-sample point evaluation rather than batched throughput, and the quadratic depth term is an empirical response-surface fit across sampled depths $d \in \{3, 6, 9\}$.

To eliminate session drift, CPU frequency fluctuations, and OS scheduling noise, all latency measurements follow a unified measurement protocol:

(i) **Pinned CPU Affinity:** The execution process is pinned to CPU Core 0 via Windows API `SetProcessAffinityMask(GetCurrentProcess(), 1)`.

(ii) **Warmup and Single-Threaded Execution:** 100 untimed warmup iterations (50 per prediction interface) precede timed execution, with XGBoost configured for single-threaded inference (`n_jobs=1`).

(iii) **Repeated Timed Evaluations:** Inference latency is timed across 1,000 single-sample evaluations using high-resolution monotonic clocks (`time.perf_counter_ns`), recording both the scikit-learn `predict` method and native `inplace_predict`. In XGBoost 3.2.0, `predict` on 2D arrays already executes in-place internally, so the measured 39–47 μs divergence reflects Python and scikit-learn wrapper overhead and array validation rather than memory copying.

(iv) **Interleaved Benchmark Runs:** During benchmarking, evaluation runs across competing algorithms are interleaved within the same pinned process session to maintain consistent CPU thermal and operating states.

3 Phase 1: Screening and Curvature Assessment

Phase 1 constructs a 2^4 full factorial design consisting of $n_F = 16$ corner runs, augmented with $n_C = 4$ center points per block ($x_1 = x_2 = x_3 = x_4 = 0$). Evaluated across all 5 seed blocks, Phase 1 comprises $N_1 = 100$ genuine training runs.

We fit a first-order response surface model with all two-factor interaction terms:

$$Y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{b=2}^{5} \gamma_b Z_b + \epsilon \tag{4}$$

where Z_b are dummy indicators for blocks 2 through 5. Table 3 details the Type III Sum of Squares ANOVA for Response Y_1.

Source of Variation	**SS**	**DF**	**MS**	**F-Stat**	***p* value**	**Partial η^2**
Intercept	7.309127	1	7.309127	2097.15	$< 10^{-15}$	0.9610
Block Effect C(block)	0.004662	4	0.001166	0.33	0.8541	0.0155
Factor A: x_1 ($\ln(\eta)$)	1.245625	1	1.245625	357.40	$< 10^{-15}$	0.8079
Factor B: x_2 (Depth)	0.082155	1	0.082155	23.57	0.0000	0.2171
Factor C: x_3 (Subsample)	0.002406	1	0.002406	0.69	0.4084	0.0081
Factor D: x_4 ($\ln(\lambda)$)	0.000089	1	0.000089	0.03	0.8731	0.0003
Interaction $x_1 \cdot x_2$	0.091725	1	0.091725	26.32	0.0000	0.2364
Interaction $x_1 \cdot x_3$	0.001878	1	0.001878	0.54	0.4649	0.0063
Interaction $x_1 \cdot x_4$	0.007452	1	0.007452	2.14	0.1474	0.0245
Interaction $x_2 \cdot x_3$	0.001106	1	0.001106	0.32	0.5747	0.0037
Interaction $x_2 \cdot x_4$	0.000092	1	0.000092	0.03	0.8714	0.0003
Interaction $x_3 \cdot x_4$	0.000081	1	0.000081	0.02	0.8790	0.0003
Residual Error	0.296247	85	0.003485	—	—	—

Table 3: Phase 1 ANOVA Table (Type III Sum of Squares) for Validation RMSE (Y_1)

Factor A ($\ln(\eta)$) dominates predictive performance ($F = 357.40$, partial $\eta^2 = 0.8079$), followed by Factor B (tree depth, $F = 23.57$) and the interaction $x_1 \cdot x_2$ ($F = 26.32, p < 10^{-6}$).

3.1 Formal Curvature Hypothesis Test

Following Montgomery (Chapter 10), curvature is tested by comparing the average response of factorial points ($\bar{y}_F$) against center points ($\bar{y}_C$):

$$\text{SS}_{\text{Curvature}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C} \tag{5}$$

Across the 100 runs evaluated in Phase 1:

$$n_F = 80, \quad \bar{y}_F = 0.6333, \quad n_C = 20, \quad \bar{y}_C = 0.4989, \quad \bar{y}_F - \bar{y}_C = 0.1343 \tag{6}$$

yielding $\mathrm{SS}_{\mathrm{Curvature}} = 0.288786$. Estimating pure error variance from center point replicates within blocks (subsampling pure error, df = 15) yields $\mathrm{MS}_{\mathrm{PE,\ Center}} = 2.74e-06$. The resulting test statistic is:

$$F_{\mathrm{Curvature}} = \frac{\mathrm{MS}_{\mathrm{Curvature}}}{\mathrm{MS}_{\mathrm{PE,\ Center}}} = 105432.31 \quad (p < 10^{-15}) \tag{7}$$

If evaluated against overall pure error pooled across all replicated design coordinates ($\mathrm{df} = 83, \mathrm{MS}_{\mathrm{PE,\ All}} = 6.98 \times 10^{-5}$), the F-statistic is also significant ($F \approx 4{,}140, p < 10^{-15}$). Under either reference, the null hypothesis of planarity is rejected, confirming curvature that requires second-order augmentation.

4 Phase 2: Central Composite Design Augmentation

4.1 Face-Centered Central Composite Design (FCCD)

To fit a second-order polynomial without discarding Phase 1 runs, the design is augmented into a Face-Centered Central Composite Design (FCCD, $\alpha = 1.0$), adding $2k = 8$ axial runs evaluated across the 5 seed blocks (40 additional runs, cumulative $N = 140$).

The second-order response surface model is:

$$\begin{aligned} Y = \beta_0 + \sum_{i=1}^{4} \beta_i x_i + \sum_{i=1}^{4} \beta_{ii} x_i^2 \\ + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{b=2}^{5} \gamma_b Z_b + \epsilon \end{aligned} \tag{8}$$

Table 4 reports the second-order ANOVA for Validation RMSE ($R^2 = 0.9951, \mathrm{Adj}\ R^2 = 0.9944$). Because residual variance varies across the factor domain, both OLS and HC3 heteroscedasticity-consistent standard errors and p-values are reported.

Source of Variation	**SS**	**DF**	**MS**	**F-Stat**	**OLS p**	**HC3 SE**	**HC3 p**
Intercept	4.718852	1	4.718852	54749.42	$< 10^{-15}$	0.0021	$< 10^{-15}$
Block Effect C(block)	0.006966	4	0.001742	20.21	< 0.0001	—	—
Factor A: x_1 ($\ln(\eta)$)	1.388673	1	1.388673	16111.76	$< 10^{-15}$	0.0012	$< 10^{-15}$
Factor B: x_2 (Depth)	0.101706	1	0.101706	1180.02	$< 10^{-15}$	0.0013	$< 10^{-15}$
Factor C: x_3 (Subsample)	0.002013	1	0.002013	23.36	< 0.0001	0.0012	0.0001
Factor D: x_4 ($\ln(\lambda)$)	0.000044	1	0.000044	0.52	0.4743	0.0012	0.5483
Quadratic x_1^2	0.146265	1	0.146265	1697.01	$< 10^{-15}$	0.0023	$< 10^{-15}$
Quadratic x_2^2	0.008779	1	0.008779	101.85	$< 10^{-15}$	0.0035	< 0.0001
Quadratic x_3^2	0.000005	1	0.000005	0.05	0.8188	0.0023	0.8005
Quadratic x_4^2	0.000006	1	0.000006	0.08	0.7844	0.0020	0.7197
Interaction $x_1 \cdot x_2$	0.091725	1	0.091725	1064.22	$< 10^{-15}$	0.0013	$< 10^{-15}$
Interaction $x_1 \cdot x_3$	0.001878	1	0.001878	21.79	< 0.0001	0.0013	0.0003
Interaction $x_1 \cdot x_4$	0.007452	1	0.007452	86.46	$< 10^{-15}$	0.0013	< 0.0001
Interaction $x_2 \cdot x_3$	0.001106	1	0.001106	12.83	0.0005	0.0013	0.0050
Interaction $x_2 \cdot x_4$	0.000092	1	0.000092	1.07	0.3040	0.0013	0.4118
Interaction $x_3 \cdot x_4$	0.000081	1	0.000081	0.94	0.3334	0.0013	0.4400
Residual Error	0.010429	121	0.000086	—	—	—	—

Table 4: Full Second-Order ANOVA Table for Validation RMSE (Y_1) with HC3 Robust Standard Errors

The ANOVA confirms substantial quadratic curvature along x_1^2 ($F = 1{,}697.01, p < 10^{-15}$; HC3 $p < 10^{-15}$) and x_2^2 ($F = 101.85, p < 10^{-15}$; HC3 $p < 0.0001$). Under HC3 covariance estimation, structural effects $x_1, x_2, x_1^2, x_2^2, x_1x_2$, and x_1x_4 remain significant ($p < 0.0001$). Table 5 provides the corresponding ANOVA for Inference Latency (Y_2).

Source of Variation	SS	DF	MS	F-Stat	p value
Intercept	555450.04	1	555450.04	1903.08	$< 10^{-15}$
Block Effect C(block)	876.94	4	219.24	0.75	0.5591
Factor A: x_1 ($\ln(\eta)$)	1265.46	1	1265.46	4.34	0.0394
Factor B: x_2 (Depth)	236841.33	1	236841.33	811.46	$< 10^{-15}$
Factor C: x_3 (Subsample)	179.04	1	179.04	0.61	0.4350
Factor D: x_4 ($\ln(\lambda)$)	483.26	1	483.26	1.66	0.2006
Quadratic x_1^2	1576.98	1	1576.98	5.40	0.0218
Quadratic x_2^2	11634.52	1	11634.52	39.86	< 0.0001
Quadratic x_3^2	8.35	1	8.35	0.03	0.8660
Quadratic x_4^2	116.71	1	116.71	0.40	0.5284
Interaction $x_1 \cdot x_2$	662.90	1	662.90	2.27	0.1344
Interaction $x_1 \cdot x_3$	14.00	1	14.00	0.05	0.8270
Interaction $x_1 \cdot x_4$	829.24	1	829.24	2.84	0.0945
Interaction $x_2 \cdot x_3$	813.68	1	813.68	2.79	0.0976
Interaction $x_2 \cdot x_4$	47.57	1	47.57	0.16	0.6871
Interaction $x_3 \cdot x_4$	52.53	1	52.53	0.18	0.6721
Residual Error	35316.19	121	291.87	—	—

Table 5: Full Second-Order ANOVA Table for Inference Latency (Y_2, μs/sample)

4.2 Lack-of-Fit Decomposition and Structural Model Adequacy

In response surface methodology [1], evaluating model adequacy requires decomposing the model residual sum of squares into pure experimental error and structural lack of fit. In our blocked central composite design ($N = 140$ runs across 5 blocks), non-center design coordinates are observed once per block. Genuine within-coordinate and within-block replication exists exclusively at the center points ($x_1 = x_2 = x_3 = x_4 = 0$), where 4 model-seed replicates within each of the 5 blocks provide exactly $\mathrm{df}_{\mathrm{PE,\ center}} = 5 \times (4 - 1) = 15$ degrees of freedom of genuine within-cell pure replicate error ($\mathrm{SS}_{\mathrm{PE,\ center}} = 0.000041$, $\mathrm{MS}_{\mathrm{PE,\ center}} = 2.74e-06$).

Fitting a saturated additive cell model containing all $m = 25$ unique design coordinates and 4 block parameters (29 parameters) leaves $140-29 = 111$ degrees of freedom with SS = 0.001579 (MS = $1.42e-05$). Rather than representing uniform replicate noise, these 111 df decompose into:

(i) Genuine center-point pure replicate error (15 df, SS = 0.000041);

(ii) Treatment × Block interaction / split-specific treatment response ($111 - 15 = 96$ df, $\mathrm{SS} = 0.001538, \mathrm{MS} = 1.60e-05$).

Evaluating the interaction against center pure error yields $F = 5.85$ ($p = 0.0002$), establishing statistically significant split-by-hyperparameter interaction: different train/validation data splits exhibit subtle variations in optimal response curvature.

The remaining $121-111 = 10$ df represent structural lack of fit ($\mathrm{SS}_{\mathrm{LoF}} = 0.008850, \mathrm{MS}_{\mathrm{LoF}} = 0.000885$). Table 6 details this complete decomposition for both responses.

Source of Variation / Model	SS	DF	MS	F	p value	Reference / RMS Misfit
Decomposition of Second-Order Residual (Y_1: Validation RMSE)						
Structural Lack of Fit	0.008850	10	0.000885	323.10	$< 10^{-15}$	vs. Center PE
Treatment × Block Interaction	0.001538	96	1.60×10^{-5}	5.85	0.0002	vs. Center PE
Genuine Center Pure Error	0.000041	15	2.74×10^{-6}	—	—	Base Replicate
Total Model Residual	0.010429	121	8.62×10^{-5}	—	—	RMS = 0.0093
Saturated Additive Baseline	0.001579	111	1.42×10^{-5}	62.21*	$< 10^{-15}$	(*LoF vs. Additive)
Restricted Domain (Y_1: $x_1 \geq -0.5$, $\eta \geq 0.033$, $N = 95$)						
Structural Lack of Fit	0.000324	2	0.000162	10.11	0.0001	RMS = 0.0127
Additive Baseline Residual	0.001203	75	1.60×10^{-5}	—	—	Saturated Base
Total Restricted Residual	0.001528	77	1.98×10^{-5}	—	—	RMS = 0.0045
Latency Residual Decomposition (Y_2: Latency, μs)						
Structural Lack of Fit	3322.37	10	332.24	1.20	0.3625	vs. Center PE (RMS = 18.23)
Treatment × Block Interaction	27844.93	96	290.05	1.05	0.4906	vs. Center PE
Genuine Center Pure Error	4148.89	15	276.59	—	—	Base Replicate
Total Model Residual	35316.19	121	291.87	1.15*	0.3305	(*LoF vs. Additive)

Table 6: Residual Variance Decomposition (121 df), Lack-of-Fit Tests, and Domain Restriction Sensitivity

For Validation RMSE (Y_1) on the full operational domain:

- Tested against center pure error (15 df), lack of fit is statistically significant ($F = 323.10, p < 10^{-15}$).
- Tested against the saturated additive baseline (111 df, pooling interaction and replicate noise), lack of fit is also significant ($F = 62.21, p = 7.78e - 41$).

Under either reference baseline, the root-mean-square misfit per degree of freedom is $\sqrt{\text{MS}_{\text{LoF}}} = 0.0297$, which is approximately 7.8× the additive residual standard deviation ($\sqrt{\text{MS}_{\text{PE}}} = 0.0038$) and 17.5× the center pure error standard deviation ($\sqrt{\text{MS}_{\text{PE, center}}} = 0.0017$). By contrast, the unweighted root-mean-square cell prediction error across the 25 unique design points is $\sqrt{\frac{1}{25}\sum_{i=1}^{25}(\bar{y}_i - \hat{y}_i)^2} \approx 0.0084$, while the observation-weighted root-mean-square misfit across all 140 design evaluations is $\sqrt{\text{SS}_{\text{LoF}}/140} \approx 0.0080$ (down-weighting non-center cells relative to the replicated center).

To inspect where this lack of fit concentrates across the hypercube, per-design-point residual contributions $n_i(\bar{y}_i - \hat{y}_i)^2$ reveal that the single largest lack-of-fit discrepancy occurs at corner $(1, 1, -1, -1)$ ($x_1 = 1, x_2 = 1, x_3 = -1, x_4 = -1$, contribution ≈ 0.00153), where high learning rate and depth combined with low subsampling and regularization challenge second-order surface adequacy. Substantial discrepancies also concentrate at low learning rates ($x_1 = -1, \eta = 0.01$), notably at $(-1, -1, 1, 1)$ (0.00097) and $(-1, 1, -1, 1)$ (0.00070), reflecting underfitting when 100 boosting iterations are paired with low step sizes. When refitting on the restricted sub-domain $x_1 \geq -0.5$ ($\eta \geq 0.033$), the sum of squares of lack of fit drops by 96.3% (from 0.008850 to 0.000324), and the RMS misfit declines by 57% to 0.0127. However, we emphasize that dropping $x_1 = -1$ leaves only two factor levels ($x_1 \in \{0, 1\}$). Because two levels induce the identity $x_1^2 \equiv x_1$, separate linear and quadratic coefficients along Factor A are unidentifiable from those levels alone, reducing df_{LoF} from 10 to 2. Consequently, this restricted test cannot assess quadratic curvature across $x_1 \in [0, 1]$ where all candidate optima reside. Even within this restricted region, lack of fit remains statistically significant ($F = 10.11, p = 0.0001$), with RMS

misfit (0.0127) remaining roughly 3× the additive residual standard deviation (0.0040). This confirms that while low-η underfitting drives the largest discrepancies, a second-order polynomial remains an approximation with non-negligible structural bias across gradient boosting loss surfaces.

For Latency (Y_2), structural lack of fit is non-significant against both reference baselines ($F = 1.20, p = 0.3625$ against center pure error; $F = 1.15, p = 0.3305$ against the saturated additive baseline). The treatment × block interaction is also non-significant ($F = 1.05, p = 0.4906$), confirming that latency is invariant across seed blocks and data splits. To characterize the functional form of latency growth, five candidate models were compared against empirical measurements, as reported in Table 7. For the log-linear specification, AIC and BIC values are adjusted via the Jacobian of transformation ($+2\sum \ln(y_i) = 1{,}450.8$) to allow direct likelihood comparison on the natural microsecond scale.

Candidate Latency Model	R^2	**Adj** R^2	**AIC** (Adjusted)	**BIC** (Adjusted)	**RMSE** (μs)
Linear (depth)	0.792	0.790	1255.2	1261.1	21.26
Quadratic (depth + depth^2)	0.859	0.857	1202.6	1211.4	17.56
Log-Linear (ln(latency) depth)	0.817	0.815	1218.1	1224.0	19.57
Multi-Factor (depth, depth^2, eta, sub, lam)	0.868	0.863	1199.5	1217.1	17.18
Full CCD Second-Order (All factors + Block)	0.882	0.864	1209.6	1265.5	17.08

Table 7: Comparison of Candidate Regression Models for Inference Latency (Y_2)

The quadratic latency model ($R^2 = 0.859$, adjusted AIC = 1202.6) yields lower AIC than the linear model ($R^2 = 0.792$, AIC = 1255.2) and the log-linear model (adjusted AIC = 1218.1), consistent with the significant x_2^2 term in ANOVA ($F = 39.86, p < 0.0001$). Latency exhibits super-linear growth across the empirical depth range ($d \in \{3, 6, 9\}$), with the quadratic fit capturing accelerating tree traversal costs while avoiding the severe over-prediction of exponential formulations. Comparing standard prediction against `inplace_predict` shows an execution gap of 39–47 μs, indicating that Python and scikit-learn wrapper overhead adds fixed baseline latency while base traversal scales with tree depth.

5 Statistical Residual Diagnostics

To evaluate regression assumptions, diagnostic checks were conducted on externally studentized residuals:

(a) **Normality:** Evaluated via Shapiro-Wilk on studentized residuals: $W = 0.9971, p = 0.9947$. The test fails to reject the null hypothesis of normality at $\alpha = 0.05$.

(b) **Variance Homogeneity across Blocks:** Assessed via Levene's test across the 5 seed blocks: $W = 0.1754, p = 0.9507$ (fails to reject equal block variance). Brown-Forsythe testing across the 25 design points yields $W = 0.6872, p = 0.8552$, failing to reject uniform replicate dispersion across design coordinates.

(c) **Reconciliation of Breusch-Pagan and Brown-Forsythe:** The Breusch-Pagan test detects continuous heteroscedasticity across the four-factor hypercube (LM = $70.10, p < 0.0001$). This does not contradict the Brown-Forsythe block homogeneity: while variance across seed blocks is uniform, residual variance compresses directionally toward regions

of higher η and deeper trees where the boosting loss reaches an asymptote. This heteroscedasticity motivates using HC3 robust standard errors for coefficient tests in Table 4; note, however, that HC3 standard errors do not correct for non-constant variance when constructing ordinary prediction intervals.

(d) **Independence and Run Order:** Durbin-Watson statistic evaluated in actual randomized execution order yields $DW = 1.9918$. The Ljung-Box test ($Q = 4.01, p = 0.5486$) and Wald-Wolfowitz Runs test ($p = 0.8576$) fail to detect serial correlation or thermal execution drift.

(e) **Influence and Leverage:** Maximum Cook's distance is $D_{\max} = 0.097$, well below the commonly cited heuristic reference cutoff of 1.0.

Figure 1 displays the 4-in-1 residual diagnostic panel.

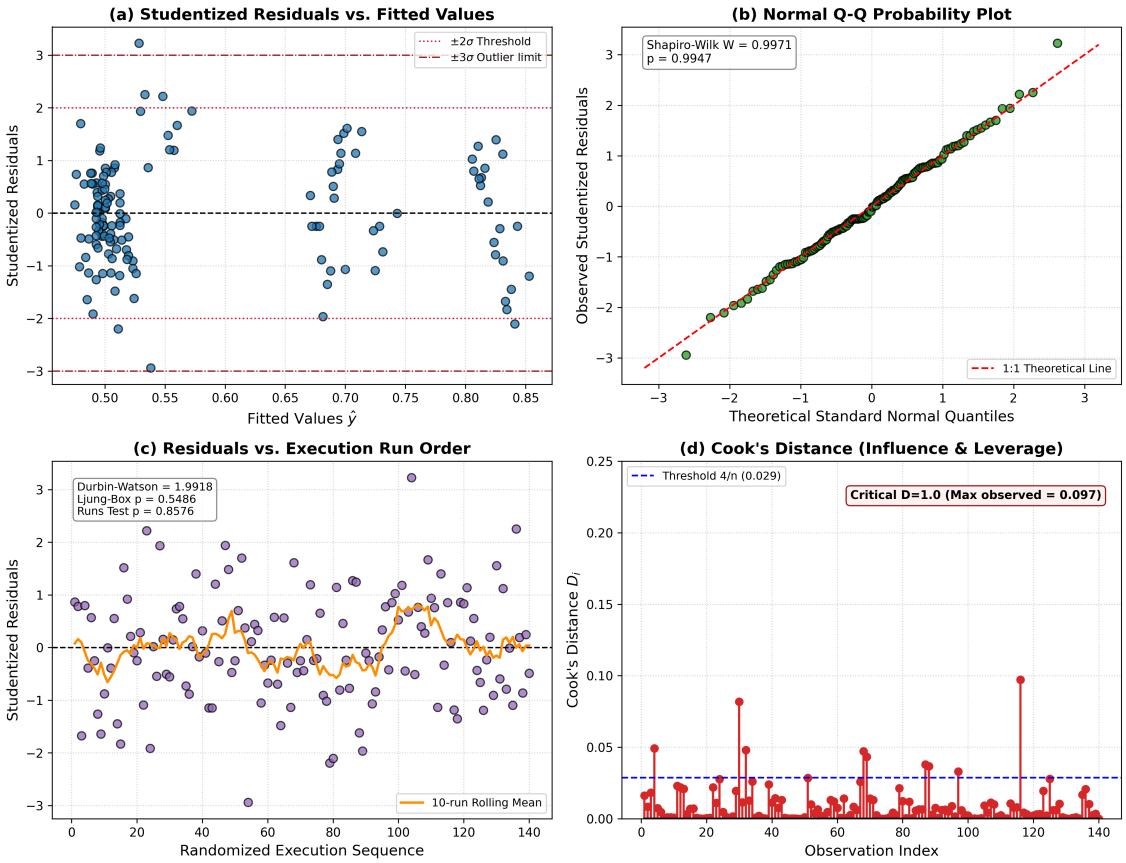


Figure 1: Residual Diagnostics Panel for Second-Order Validation RMSE Model: (a) Studentized residuals vs. fitted values; (b) Normal Q-Q probability plot ($p = 0.9947$); (c) Residuals vs. randomized run execution order ($DW = 1.9918$, Ljung-Box $p = 0.5486$, Runs test $p = 0.8576$); (d) Cook's distance influence plot ($D_{\max} = 0.097$, reference guideline 1.0).

6 Phase 3: Canonical and Ridge Analysis

6.1 Matrix Representation and Stationary Point

The fitted second-order model is expressed in matrix notation:

$$\hat{y} = b_0 + \mathbf{x}^T\mathbf{b} + \mathbf{x}^T\mathbf{B}\mathbf{x} \tag{9}$$

where $b_0 = 0.4994$ (averaging block effects equally), $\mathbf{b} = [-0.1242, -0.0336, -0.0047, -0.0007]^T$, and:

$$\mathbf{B} = \begin{bmatrix} 0.1065 & 0.0169 & -0.0024 & -0.0048 \\ 0.0169 & 0.0261 & -0.0019 & -0.0005 \\ -0.0024 & -0.0019 & 0.0006 & 0.0005 \\ -0.0048 & -0.0005 & 0.0005 & 0.0007 \end{bmatrix} \tag{10}$$

The unconstrained stationary point $\mathbf{x}_0$ satisfies $\mathbf{b} + 2\mathbf{B}\mathbf{x}_0 = \mathbf{0}$:

$$\mathbf{x}_0 = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b} = [0.4171, 1.3524, 15.6944, -6.8009]^T \tag{11}$$

The stationary coordinate along Factor C ($x_3 = 15.6944$) lies far outside the operational cube $[-1, +1]^4$. Its Euclidean distance from the coded origin is $\|\mathbf{x}_0\| = 17.16$, while its Euclidean distance from the boundary of the feasible hypercube is 15.80 coded units. The unconstrained stationary point has a predicted response of $\hat{y}_0 = 0.4161$, an extrapolation artifact indicating that unconstrained quadratic optimization yields an infeasible point.

6.2 Spectral Decomposition and Ridge Analysis

Eigenvalues of $\mathbf{B}$ are $\lambda_1 = 0.000057, \lambda_2 = 0.000873, \lambda_3 = 0.022759, \lambda_4 = 0.110196$ ($\sum \lambda_i = \text{trace}(\mathbf{B}) = 0.133886$). While point estimates are positive, Rademacher wild bootstrap (2,000 replications) reveals that the 95% bootstrap interval for λ_1 spans $[-0.0039, 0.0022]$, with 68.2% of bootstrap samples yielding $\min(\lambda) \leq 0$. The canonical eigenvectors (columns of the orthogonal decomposition matrix $\mathbf{M}$) show that the smallest eigenvalue λ_1 aligns along $\mathbf{v}_1 = [0.023, -0.053, -0.724, 0.687]^T$, representing a contrast between Factor C (subsampling, -0.724) and Factor D (L_2 regularization, $+0.687$), while λ_2 corresponds to $\mathbf{v}_2 = [-0.043, -0.037, -0.687, -0.725]^T$, representing a joint direction along subsampling and regularization. By contrast, the dominant quadratic curvature axes align along Factor B (depth, $\mathbf{v}_3 = [-0.198, 0.978, -0.060, 0.018]^T$, $\lambda_3 = 0.022759$) and Factor A (learning rate, $\mathbf{v}_4 = [-0.979, -0.198, 0.025, 0.044]^T$, $\lambda_4 = 0.110196$). The system is classified as a **stationary or rising ridge**, with curvature along x_3 and x_4 statistically consistent with zero.

Constrained optimization over the hypercube $[-1, 1]^4$ with integer depth $d \in \{3, \ldots, 9\}$ yields the results in Table 8.

Depth	**Coded x_1**	**Coded x_2**	**Coded x_3**	**Coded x_4**	**Optimal η**	**Predicted RMSE**	**SE ($\hat{y}$)**
3	0.8103	−1.0000	1.00	1.00	0.2173	0.4909	0.0032
4	0.7573	−0.6667	1.00	1.00	0.1985	0.4724	0.0030
5	0.7043	−0.3333	1.00	1.00	0.1814	0.4592	0.0032
6	0.6513	+0.0000	1.00	1.00	0.1658	0.4511	0.0033
7	0.5983	+0.3333	1.00	1.00	0.1515	0.4483	0.0032
8	0.5453	+0.6667	1.00	1.00	0.1384	0.4506	0.0031
9	0.4923	+1.0000	1.00	1.00	0.1265	0.4582	0.0034

Table 8: Best Predicted Validation RMSE at Each Discrete Tree Depth

The constrained single-objective optimum within the hypercube occurs at depth 7 ($\mathbf{x}^*_{\text{SO}}$: $x_1 = 0.5983, x_2 = 0.3333, x_3 = 1.0000, x_4 = 1.0000$) with predicted RMSE 0.4483. Figures 2 and 3 display the response surfaces.

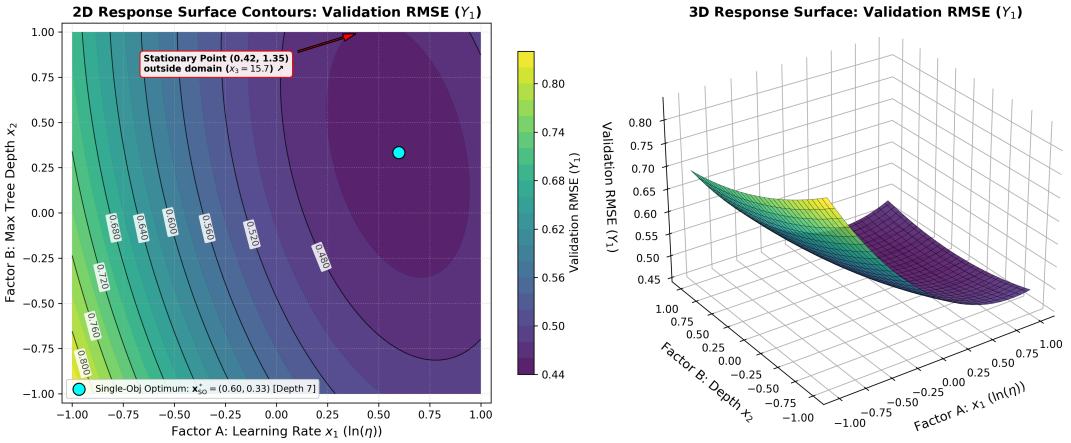


Figure 2: Response Surface Topography for Validation RMSE (Y_1): (a) 2D contour slice across Factor A ($\ln(\eta)$) and Factor B (depth), passing through the constrained optimum ($x_3 = 1, x_4 = 1$). Arrow indicates trajectory toward unconstrained stationary point outside domain; (b) 3D wireframe projection.

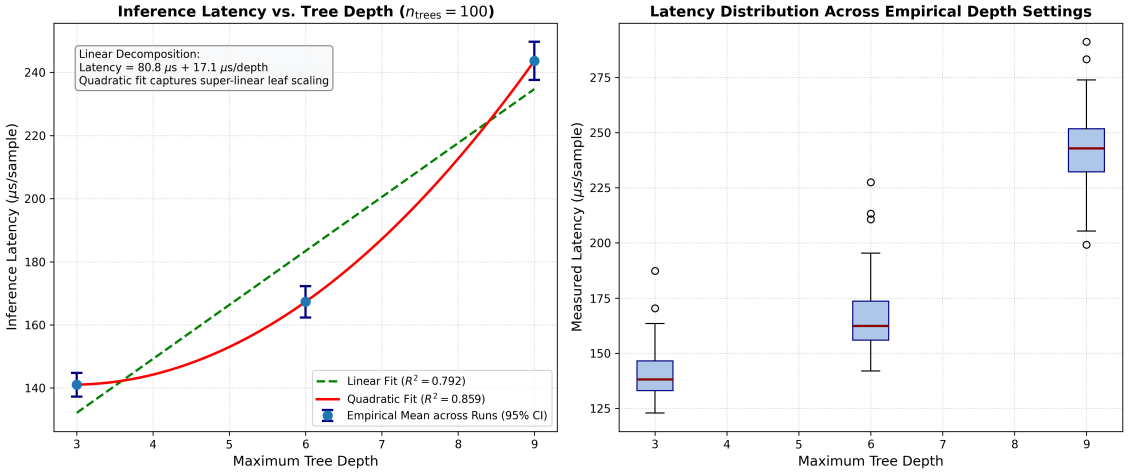


Figure 3: Inference Latency Modelling: (a) Empirical mean latency with 95% CI error bars vs. linear ($R^2 = 0.792$) and quadratic ($R^2 = 0.859$) fits. Note the linear base intercept (≈ 80.8 μs) arising from Python execution overhead; (b) Distribution boxplots across discrete depths.

7 Phase 4: Multi-Objective Desirability Optimization

Minimizing predictive error requires deeper trees, which increases inference latency. To balance predictive error against inference latency, we use Derringer-Suich desirability functions [6]:

$$d_r(Y_r) = \begin{cases} 1.0, & Y_r < L_r \\ \left(\frac{U_r - Y_r}{U_r - L_r}\right)^{s_r}, & L_r \leq Y_r \leq U_r \\ 0.0, & Y_r > U_r \end{cases} \tag{12}$$

The composite desirability D is formulated with explicit importance weights (w_1, w_2):

$$D = (d_1(Y_1)^{w_1} \cdot d_2(Y_2)^{w_2})^{\frac{1}{w_1 + w_2}} \tag{13}$$

In the baseline scenario, equal weights $w_1 = 1, w_2 = 1$ are used with engineering limits $L_1 = 0.450, U_1 = 0.700$ for RMSE and $L_2 = 100.0, U_2 = 180.0$ μs for latency ($s_1 = s_2 = 1.0$).

To select a deployable operating point, desirability D was evaluated across a high-resolution factor grid ($21 \times 7 \times 2 \times 2 = 588$ design coordinates) with discrete depths $d \in \{3, \ldots, 9\}$. While unconstrained maximization pushes toward the upper boundary $\eta \to 0.30$ ($x_1 = 1.0$, reaching $D = 0.6952$) due to the 100-tree ceiling, operating directly on boundary edges risks boundary extrapolation. Holding depth $d = 4$, subsample 1.0, and $\lambda = 0.830$, composite desirability varies smoothly across the interior: $D = 0.6662$ at $\eta = 0.20$, $D = 0.6782$ at $\eta = 0.2324$, $D = 0.6833$ at $\eta = 0.25$, and $D = 0.6943$ at $\eta = 0.30$. We chose the interior coordinate at $x_1 = 0.8500$ ($\eta = 0.2324$) as the selected operational multi-objective compromise configuration $\mathbf{x}^*_{\text{MO}}$:

$$\begin{aligned} \mathbf{x}^*_{\text{MO}} &= [0.8500, -0.6667, 1.0000, -0.0812]^T \\ &\Longrightarrow (\eta = 0.2324, \text{depth} = 4, \text{subsample} = 1.0000, \lambda = 0.8295) \end{aligned} \tag{14}$$

achieving $D = 0.6782$ ($d_1 = 0.8783, d_2 = 0.5236$). Predicted metrics at $\mathbf{x}^*_{\text{MO}}$ are $\hat{Y}_1 = 0.4804$ and $\hat{Y}_2 = 138.11$ μs.

Table 9 reports sensitivity across alternative specification limits and weights. Across all scenarios, the optimal depth clusters between 3 and 5, indicating that shallow-to-moderate tree depths provide the best balance between accuracy and latency.

Scenario	**RMSE Bounds** $[L_1, U_1]$	**Latency Bounds** $[L_2, U_2]$	**Weights** (w_1, w_2)	**Opt Depth**	**Opt** η	**Best** D
Standard Config	[0.45, 0.70]	[100, 180]	(1, 1)	4	0.3000	0.6952
Strict Latency	[0.45, 0.70]	[90, 140]	(1, 2)	3	0.3000	0.2206
Strict Accuracy	[0.44, 0.60]	[100, 200]	(2, 1)	5	0.2135	0.7293
Narrow Range	[0.46, 0.65]	[110, 160]	(1, 1)	4	0.3000	0.6597

Table 9: Desirability Optimization Sensitivity Analysis Across Bounds and Weights

Figure 4 illustrates the historical DOE development design space, showing individual block run realizations as background observations, configuration-level 5-block means with error bars, the configuration-mean Pareto front, and the operational coordinates $\mathbf{x}^*_{\text{MO}}$ and $\mathbf{x}^*_{\text{SO}}$.

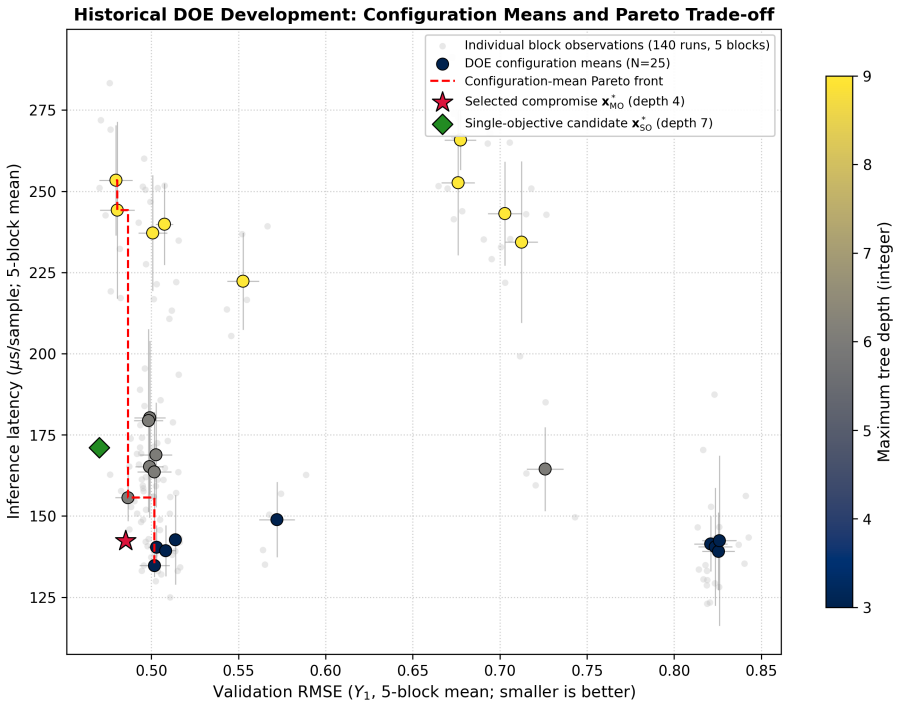


Figure 4: Historical DOE Multi-Objective Trade-Off: Validation RMSE vs. Inference Latency across 5 Seed Blocks. Light gray points show individual block run observations (140 runs across 5 blocks). Filled markers show the 25 unique design configuration means (averaged over blocks) colored by tree depth, with error bars indicating ± 1 SD across blocks. The red dashed line traces the configuration-mean Pareto front. Highlighted operating coordinates include the multi-objective compromise $\mathbf{x}^*_{\text{MO}}$ (depth 4, red star), single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (depth 7, green diamond), and empirical confirmation means ($m = 10$ seeds).

8 Phase 5: Empirical Confirmation and Benchmark Comparison

8.1 Confirmation Experiments at Multi-Objective and Single-Objective Optima

To test surrogate fidelity, $m = 10$ confirmation trials were executed at $\mathbf{x}^*_{\text{MO}}$ across fresh seeds disjoint from the design ($\mathcal{S}_{\text{conf}} = \{505, 606, 707, 808, 909, 1010, 1111, 1212, 1313, 1414\}$). Because fresh seeds introduce block variance (σ^2_{block} estimated with only 4 df) alongside model parameter covariance (MS_E with 121 df), treating block variance uncertainty as negligible would understate interval width. The total theoretical variance for the confirmation mean $\bar{y}_m$ at coordinate $\mathbf{x}_0$ with leverage $h_0 = \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$ is:

$$V_{\text{conf}} = \sigma^2_E \left(\frac{1}{m} + h_0\right) + \sigma^2_{\text{block}} \left(\frac{1}{m} + \frac{1}{5}\right) \tag{15}$$

Under the randomized block design with $a = 28$ runs per block, the expected mean squares satisfy $\mathbb{E}[\text{MS}_E] = \sigma^2_E$ and $\mathbb{E}[\text{MS}_{\text{block}}] = \sigma^2_E + 28\,\sigma^2_{\text{block}}$, giving the unbiased ANOVA estimator $\hat{\sigma}^2_{\text{block}} = (\text{MS}_{\text{block}} - \text{MS}_E)/28$. Substituting $\hat{\sigma}^2_{\text{block}}$ and $\hat{\sigma}^2_E = \text{MS}_E$ into V_{conf} yields the linear combination of independent mean squares required by Satterthwaite's synthesis:

$$\hat{V}_{\text{conf}} = \text{MS}_E \left(\frac{1}{m} + h_0\right) + \frac{\text{MS}_{\text{block}} - \text{MS}_E}{28} \left(\frac{1}{m} + \frac{1}{5}\right) = c_1 \text{MS}_E + c_2 \text{MS}_{\text{block}} \tag{16}$$

where $c_2 = \frac{1/m+1/5}{28}$ and $c_1 = (1/m + h_0) - c_2$. Propagating variance-component degrees of freedom via Satterthwaite's approximation yields effective degrees of freedom:

$$\nu_{\text{eff}} = \frac{\hat{V}_{\text{conf}}^2}{\frac{(c_1 \text{MS}_E)^2}{121} + \frac{(c_2 \text{MS}_{\text{block}})^2}{4}} \tag{17}$$

For $\mathbf{x}^*_{\text{MO}}$ ($h_0 = 0.1043$), Satterthwaite effective degrees of freedom evaluate to $\nu_{\text{eff}} = 14.0$ ($t_{0.025,14.0} = 2.145$), widening the 95% prediction interval to $[0.4677, 0.4932]$ (compared to $[0.4687, 0.4922]$ under the unadjusted $t_{0.025,121}$ approximation). For the depth-7 single-objective coordinate $\mathbf{x}^*_{\text{SO}}$ ($h_0 = 0.1210$), $\nu_{\text{eff}} = 15.1$ ($t_{0.025,15.1} = 2.131$), producing $[0.4354, 0.4612]$. We note that Satterthwaite prediction intervals pool structural lack of fit (10 df) and treatment $\times$ block interaction (96 df) into MS_E, assuming homogeneous residual variance across design cells without safeguarding against local polynomial bias. Table 10 compares predictions against empirical results.

Configuration / Response Metric	**Surrogate Pred ($\hat{y}$)**	**95% Pred Interval (PI)**	**Empirical Mean $\pm$ SD**	**Confirmation Status**
DOE Multi-Objective Optimum $\mathbf{x}^*_{\text{MO}}$ (Depth 4)				
Validation RMSE (Y_1)	0.4804	$[0.4677, 0.4932]$	0.4853 ± 0.0115	**Pass** (Inside 95% PI)
Holdout Test RMSE	—	—	0.4888 ± 0.0032	Holdout Test Set
Inference Latency (μs)	138.1	$[122.8, 153.4]$	142.3 ± 7.8	**Pass** (Inside 95% PI)
DOE Single-Objective Candidate $\mathbf{x}^*_{\text{SO}}$ (Depth 7)				
Validation RMSE (Y_1)	0.4483	$[0.4354, 0.4612]$	0.4700 ± 0.0093	**Not Confirmed** (Optimism: +0.0217)
Holdout Test RMSE	—	—	0.4691 ± 0.0034	Holdout Test Set
Inference Latency (μs)	187.0	$[171.1, 202.9]$	171.1 ± 10.9	Borderline (At Lower PI)

Table 10: Model Prediction vs. Empirical Confirmation Trials ($m = 10$ Fresh Seeds)

For the multi-objective compromise coordinate $\mathbf{x}^*_{\text{MO}}$ (depth 4), empirical validation RMSE of 0.4853 ± 0.0115 falls directly inside the Satterthwaite prediction interval ($[0.4677, 0.4932]$), providing pointwise confirmation consistent with the surrogate. Empirical latency was $142.33 \pm 7.79\,\mu$s, also falling within $[122.8, 153.4]\,\mu$s. Test RMSE on the external holdout set ($N = 4{,}128$) averaged 0.4888 ± 0.0032. Pointwise confirmation at $\mathbf{x}^*_{\text{MO}}$ does not imply global surrogate accuracy across the entire hypercube.

For the single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (depth 7), the surrogate predicted $\hat{y} = 0.4483$ with Satterthwaite 95% PI $[0.4354, 0.4612]$, while empirical confirmation yielded 0.4700 ± 0.0093. Because empirical validation RMSE lies 0.0088 above the upper bound of the Satterthwaite interval, the prediction was not confirmed. The surrogate exhibited structural optimism of $+0.0217$ ($6.8 \times \text{SE}(\hat{y})$), indicating that quadratic interpolation between sampled design depths $\{3, 6, 9\}$ does not adequately track the diminishing returns of deeper trees. Measured inference latency ($171.1 \pm 10.9\,\mu$s) sits directly on the lower prediction interval bound ($171.1\,\mu$s), representing a borderline confirmation. On the holdout test set, depth 7 achieves an external test RMSE of 0.4691 ± 0.0034, competitive with Bayesian baselines.

8.2 Benchmark Evaluation Against Random Search and Optuna TPE

To rigorously compare the classical Design of Experiments framework against state-of-the-art heuristic search algorithms, we conducted prospective benchmark experiments under strictly equal evaluation budgets (140 function evaluations per search). We evaluated four baseline optimization algorithms alongside repeated DOE executions across $N = 20$ independent search replicates:

1. **Unguided Random Search:** 20 independent replicates of 140 evaluations uniformly sampled across the 4-factor domain.

2. **Single-Objective TPE:** 20 independent replicates of 140 evaluations optimizing validation RMSE via Tree-structured Parzen Estimators [4, 5].

3. **Constrained TPE:** Single-objective TPE with an operational inference latency constraint ($\leq 145\,\mu$s) evaluated via single-threaded single-sample prediction latency measured online during trial sampling.

4. **Multi-Objective TPE:** Evaluates validation RMSE and online single-sample prediction latency simultaneously to construct a candidate Pareto front, followed by post-hoc operating coordinate selection via Derringer-Suich desirability.

5. **Repeated DOE Formulations:** Pre-planned Face-Centered Central Composite Designs executed across the 5 seed blocks, evaluating both the multi-objective compromise $\mathbf{x}^*_{\text{MO}}$ (depth 4) and the single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (depth 7).

Across the six methods, winning hyperparameter configurations from the 20 search replicates (122 selection records representing 95 distinct configuration hashes, yielding 2,440 final evaluation rows) were serialized into an immutable manifest (`finalized_selections.json`, verified by SHA-256) prior to holdout evaluation. Across the prospective benchmark campaign, exactly 17,077 XGBoost model fits and 1,640,810 timed prediction evaluations were executed.

Crucially, we clarify the history of the holdout test set: the historical $v1.0.0$ experiments recorded test results during development, whereas the revised Revision-v2 pipeline programmatically prevented access to those test labels during all search iterations, surrogate fitting, and desirability calculations until winning configurations were frozen in `finalized_selections.json`. While these safeguards enforce pipeline isolation during the revised experiments, they cannot undo earlier historical exposure of the same dataset. Each frozen hyperparameter configuration was then retrained from scratch and evaluated across 20 fresh seeds (seeds 2001–2020, $\mathcal{S}_{\text{eval}}$) on the external holdout test set under dedicated single-threaded latency profiling. Single-sample inference latency was timed across 1,000 single-sample evaluations per seed using high-resolution monotonic clocks (`time.perf_counter_ns`) with CPU core affinity pinned to CPU Core 0. Regarding historical comparisons, we note that both historical confirmation and benchmark scripts applied Win32 CPU pinning, and the historical latency discrepancy between single-run snapshots and prospective runs cannot be conclusively attributed to unpinned execution or an identified environmental mechanism.

Table 11 summarizes the benchmark comparison, reporting Panel A (the prospective 20-replicate empirical evidence with between-search standard deviations, retraining standard deviations, and benchmark constraint feasibility) alongside Panel B (the historical $v1.0.0$ baseline snapshot).

Optimization Method	**Search Basis** (N Reps)	**Val RMSE** Mean ± SD	**Holdout Test RMSE** Mean ± SD	**Retrain** $\boldsymbol{\sigma}_{\text{eval}}$	**Latency** (μs ± SD)	**Feasible** ($\leq 145\,\mu$s)
Panel A: Prospective Revision-v2 Full Benchmark ($N = 20$ Search Replicates)						
Repeated DOE MO ($\mathbf{x}^*_{\text{MO}}$)	RSM (N=20)	0.4853 ± 0.0055	0.4900 ± 0.0073	0.0033	122.3 ± 2.7	20/20
Multi-Objective TPE	Parzen (N=20)	0.4881 ± 0.0093	0.4926 ± 0.0108	0.0036	121.5 ± 3.8	20/20
Constrained TPE ($\leq 145\,\mu$s)	Parzen (N=20)	0.4700 ± 0.0022	0.4704 ± 0.0032	0.0034	144.3 ± 7.1	9/20
Repeated DOE SO (d=7)	RSM (N=20)	0.4687 ± 0.0000	0.4686 ± 0.0000	0.0030	150.7 ± 0.7	0/20
Single-Objective TPE	Parzen (N=20)	0.4675 ± 0.0008	0.4669 ± 0.0012	0.0031	188.9 ± 12.1	0/20
Unguided Random Search	Uniform (N=20)	0.4687 ± 0.0014	0.4695 ± 0.0021	0.0030	181.1 ± 19.0	0/20
Panel B: Historical Baseline Snapshot (v1.0.0, Single Search Replicate)						
Historical DOE MO ($\mathbf{x}^*_{\text{MO}}$)	RSM (Single)	0.4821	0.4884	—	119.8	Yes
Historical Multi-Obj TPE	Parzen (Single)	0.4812	0.4870	—	118.8	Yes
Historical Constrained TPE	Parzen (Single)	0.4783	0.4823	—	125.4	Yes
Historical DOE SO (d=7)	RSM (Single)	0.4687	0.4690	—	147.8	No
Historical Bayesian TPE	Parzen (Single)	0.4735	0.4726	—	193.2	No
Historical Random Search	Uniform (Single)	0.4696	0.4707	—	198.1	No

Table 11: Empirical Benchmark Comparison across 20 Fresh Seeds on Fixed Holdout Test Set. Panel A reports prospective Revision-v2 evidence across $N = 20$ independent search replicates (140 fits per replicate) evaluated across 20 retraining seeds $\mathcal{S}_{\text{eval}}$ on the external holdout set ($N = 4,128$). Panel B preserves the historical v1.0.0 single-replicate baseline snapshot for reference.

Table 12 details the empirical Pareto hypervolume comparisons across candidate search frontiers and the holdout evaluated selections.

Frontier / Evaluation Protocol	Candidate Pool (N)	Non-Dom. Points	HV at $[0.60, 250]$ (Mean $\pm$ SD)	HV at $[0.65, 275]$ (Mean $\pm$ SD)
Panel A: Development Candidate Frontiers (Distinct Evaluation Protocols Noted per Row)				
Multi-Objective TPE Candidates (Single Split 42, 30-Call Search)	140×20	9.35 ± 2.3	18.3834 ± 0.3301	30.0535 ± 0.4123
Repeated DOE Candidate Fronts (5-Block Means Across Replicates)	25×20	6.10 ± 1.4	15.6128 ± 0.7268	26.6847 ± 0.8786
Fixed Full DOE Evaluated Front (Single Split 42, 27 Configs)	27	4	16.8585	28.1430
Historical DOE 2-Point Set (Single Split 42, $\mathbf{x}^*_{\mathrm{MO}}$ & $\mathbf{x}^*_{\mathrm{SO}}$)	2	2	16.0363	26.9168
Difference: Fixed Full DOE Front − MO-TPE Mean (Split 42)	—	—	-1.5249 ($p < 0.0001$)	-1.9105 ($p < 0.0001$)
Panel B: Holdout Non-Dominated Set (122 Records, 95 Configs, 20 Seeds)				
Non-Dominated Set (12 Distinct Configs)	122	12	17.6714	28.9900
⌞ Repeated DOE MO Selections	20	3	—	—
⌞ Multi-Objective TPE Selections	20	3	—	—
⌞ Constrained TPE Selections	20	5	—	—
⌞ Single-Objective TPE Selections	20	1	—	—

Table 12: Pareto Hypervolume Analysis Across Development Frontiers and External Holdout Evaluated Selections. Panel A summarizes development candidate fronts across their respective evaluation protocols (MO-TPE candidates on single development split 42 under 30-call timing, Repeated DOE candidates averaged across five seed blocks per replicate, and the fixed 27-point DOE evaluated frontier on development split 42). Panel B reports the non-dominated set among the 122 frozen evaluated selection records (95 distinct configurations) across 20 retraining seeds on the external holdout test set.

Figure 5 contrasts development candidate frontiers on development split 42 (left) with genuine holdout test set evaluations of frozen selections across 20 retraining seeds (right).

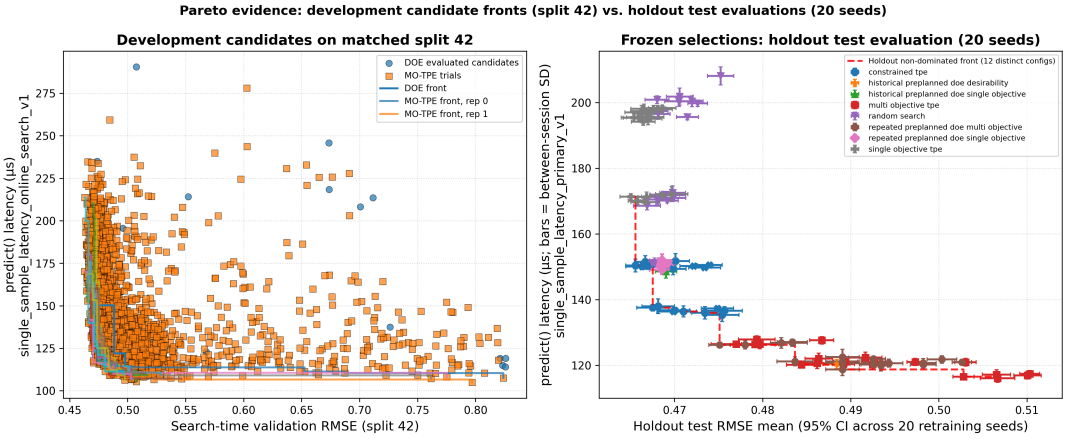


Figure 5: Pareto Evidence Architecture: (Left) Development candidate Pareto fronts on matched split 42 under the online 30-call latency protocol, showing evaluated DOE points, MO-TPE search trial clouds, and replicate-level candidate Pareto steps; (Right) Evaluated frozen selections on the external holdout test set across 20 retraining seeds, with horizontal error bars showing 95% confidence intervals for holdout test RMSE and vertical error bars showing between-session latency standard deviations. The red dashed line traces the empirical holdout non-dominated front comprising 12 distinct configurations.

Figure 6 illustrates optimization sample efficiency across training evaluations.

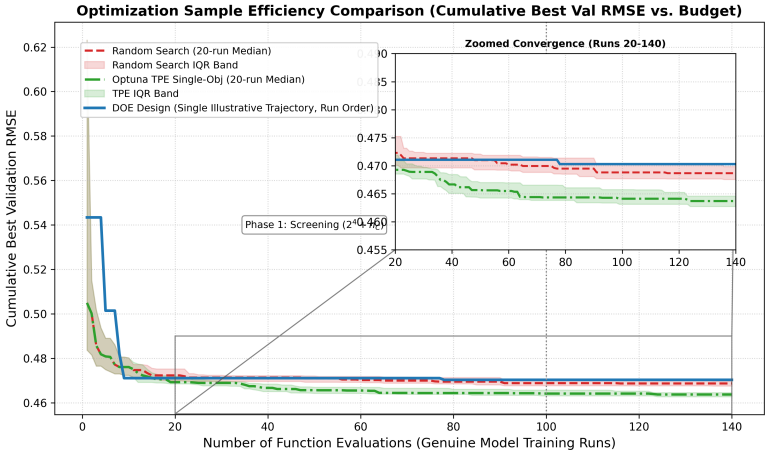


Figure 6: Sample Efficiency Convergence: Cumulative best validation RMSE vs. evaluation budget. Curves for Random Search and TPE show median trajectories ± IQR shaded bands across 20 independent runs. The DOE curve shows cumulative best in actual randomized run execution order across the pre-planned blocked design. Inset highlights fine-grained convergence between runs 20 and 140.

The prospective empirical benchmarks establish five key scientific findings:

1. **Single-Objective Predictive Performance:** Single-Objective TPE achieves the numerically lowest observed mean Test RMSE (0.46691 ± 0.00116), selecting deep trees ($d \in \{7, 8, 9\}$) with an average inference latency of $188.93 \pm 12.07\,\mu$s (inplace $157.14\,\mu$s). Neither Single-Objective TPE nor Repeated DOE Single-Objective satisfies the $\leq 145\,\mu$s latency constraint (0/20 feasible). Repeated DOE Single-Objective ($\mathbf{x}^*_{\text{SO}}$, depth 7) yields deterministic selection across replicates (between-search SD = 0.00000, retraining $\sigma = 0.00304$), achieving a mean Test RMSE of 0.46859 at $150.74 \pm 0.72\,\mu$s latency (20.2% lower latency than SO-TPE).

2. **Multi-Objective Performance Comparison:** Comparing multi-objective solutions, Repeated DOE ($\mathbf{x}^*_{\text{MO}}$, depth 4) achieves a mean Test RMSE of 0.48997 ± 0.00732 at $122.29 \pm 2.70\,\mu$s latency (inplace $89.73\,\mu$s), closely matching Multi-Objective TPE which achieves 0.49255 ± 0.01080 at $121.46 \pm 3.75\,\mu$s (inplace $88.84\,\mu$s). The difference of -0.00258 RMSE is statistically non-significant under both Welch's t-test ($t = -0.8849, p = 0.3826$) and paired t-testing across search replicates ($t = -0.966, p = 0.3463$). Both multi-objective methods achieve 100% benchmark constraint feasibility (20/20).

3. **Development Candidate Hypervolume Comparison:** On development split 42, MO-TPE candidate fronts attain a mean hypervolume of 18.3834 ± 0.3301 (reference $[0.60, 250.0]$), significantly exceeding the full evaluated DOE candidate frontier (16.8585, difference -1.5249, $p < 0.0001$). Under alternative reference $[0.65, 275.0]$, MO-TPE achieves 30.0535 ± 0.4123 versus 28.1430 for DOE (difference -1.9105, $p < 0.0001$). This evidence demonstrates a higher observed candidate hypervolume for MO-TPE on the evaluated development split under the 30-call search latency protocol, but does not establish universal algorithmic superiority across unconstrained domains or unseen data splits.

4. **Holdout Non-Dominated Set Architecture:** When evaluating the 122 frozen selected configurations (representing 95 distinct configuration hashes) across 20 fresh seeds on the external holdout test set, exactly 12 configurations form the empirical non-dominated set (hypervolume 17.6714 at reference $[0.60, 250.0]$ and 28.9900 at $[0.65, 275.0]$). We emphasize

that this is the non-dominated set *among the evaluated frozen configurations*, not the global Pareto frontier across the entire unconstrained hyperparameter domain. The set is formed by a complementary mixture of configurations: 3 Repeated DOE MO selections, 3 MO-TPE selections, 5 Constrained TPE selections, and 1 SO-TPE selection.

5. **Constrained TPE Feasibility Dynamics:** Constrained TPE achieves 20/20 feasibility during search under online 30-call timing, but yields 9/20 (45%) feasibility under the benchmark evaluation protocol ($136.70 \pm 0.82\,\mu$s for depth 6 selections, 9/9 feasible; $150.45 \pm 0.67\,\mu$s for depth 7 selections, 0/11 feasible). Plausible explanations include noisy 30-call search measurements near the boundary, protocol differences (30 calls without warmup vs. 1,000 calls with warmup), and optimizer selection effects, rather than a conclusively proven single cause.

9 Discussion and Limitations

The empirical findings highlight the complementary strengths and limitations of classical Design of Experiments and sequential Bayesian optimization:

1. **Parametric Attribution vs. Heuristic Search:** DOE provides structured variance decomposition, formal curvature testing ($F = 105432.31$), lack-of-fit testing ($F = 62.21$), and deterministic selection stability (between-search SD = 0.00000). By contrast, heuristic search algorithms treat the objective as an opaque oracle and provide no parametric insight into factor interactions.

2. **Stochastic Seed Blocking:** Random complete block design isolated an estimated $\sigma_{\text{block}} \approx 0.0077$ RMSE in unexplained residual variance (ICC = 0.4069). While primary factors account for $> 90\%$ of total sum of squares, this nuisance variation is comparable in magnitude to the differences separating competitive configurations, demonstrating the value of blocking in shielding hyperparameter evaluations from data-partition noise.

3. **Structural Lack of Fit and Model Optimism:** Second-order polynomials cannot fully capture the asymptotic convergence of gradient boosting loss. This structural limitation led to significant lack of fit ($F = 62.21, p = 7.78e-41$) with an RMS misfit of 0.0297, causing surrogate optimism at intermediate depth 7 (bias = +0.0217). Satterthwaite prediction intervals pool structural lack of fit into MS_E, widening intervals under unexplained variance but failing to protect against localized polynomial bias. Polynomial surrogates should therefore be treated as local guidance maps rather than globally precise response estimators.

4. **Complementary Search Capabilities:** While DOE excels at structured exploration within a pre-planned budget, adaptive Bayesian methods like TPE can explore arbitrary interior regions and identify slightly better accuracy points (e.g., SO-TPE test RMSE 0.46691) without polynomial bias. In multi-objective mode, however, DOE MO and MO-TPE achieve comparable test RMSE ($p = 0.38$) at similar inference latency ($\approx 122\,\mu$s).

5. **Latency Modeling and Measurement Protocol:** Single-sample inference latency follows super-linear growth across discrete depths ($R^2 = 0.859$). Dedicated single-threaded profiling with CPU pinning minimizes thermal and scheduling noise, but online search-time timing remains subject to microarchitectural jitter, emphasizing the need for rigorous post-hoc validation.

6. **Scope and Operational Context:** These conclusions are established for XGBoost with 100 boosting trees on the California Housing dataset across four hyperparameters. In higher dimensions ($D > 10$), fractional factorial screening or hybrid Bayesian-DOE strategies are warranted.

10 Reproducibility Statement

All experimental code, configuration manifests, raw execution logs, and LaTeX source files are publicly available under open-source version control at `https://github.com/ChristosHussein/xgboost-doe-rsm` (`branch codex/scientific-revision`, `PR #2`). The repository preserves the immutable historical baseline (tag `v1.0.0`) alongside the complete prospective revision campaign (17,077 model fits and 1,640,810 timed prediction inferences). Experiments were executed on an AMD Ryzen 5 9600X processor (12 logical CPUs) under `Windows-11-10.0.26300-SP0` using Python 3.11.9, XGBoost 3.2.0, Scikit-Learn 1.8.0, Statsmodels 0.15.0, Scipy 1.17.1, and Optuna 5.0.0. Execution scripts pin CPU core affinity (`CPU 0`) and fix test partitions; single-sample latency measurements nonetheless remain subject to operating system scheduling and hardware state.

11 Conclusion

This study conducted a comprehensive scientific evaluation of classical Design of Experiments (DOE) principles for hyperparameter optimization in gradient boosted decision trees. Through sequential factorial screening, central composite design augmentation, canonical spectral analysis, and desirability optimization, the framework isolated stochastic seed variance ($\sigma_{\text{block}} \approx 0.0077$), detected response curvature, and identified a multi-objective operating coordinate ($\mathbf{x}^*_{\text{MO}}$) balancing predictive accuracy with inference latency. Empirical confirmation across fresh seeds confirmed surrogate prediction intervals at $\mathbf{x}^*_{\text{MO}}$, while revealing structural polynomial optimism at the single-objective candidate $\mathbf{x}^*_{\text{SO}}$ (+0.0217 RMSE bias).

Prospective benchmark evaluations encompassing 17,077 model fits across 20 independent search replicates and 20 retraining seeds on an external holdout test set demonstrated the complementary nature of DOE and Bayesian optimization: Single-Objective TPE achieved the numerically lowest observed mean Test RMSE (0.46691 ± 0.00116) at higher latency ($188.93\,\mu$s), whereas Repeated DOE Single-Objective achieved deterministic selection (0.46859) with 20.2% lower latency ($150.74\,\mu$s). In multi-objective optimization, Repeated DOE ($\mathbf{x}^*_{\text{MO}}$) and Multi-Objective TPE showed no statistically significant difference in mean Test RMSE (0.48997 vs. 0.49255, $p = 0.38$) at comparable single-sample latency ($122.3\,\mu$s vs. $121.5\,\mu$s). On the development split, MO-TPE attained a higher candidate hypervolume (18.3834 vs. 16.8585, $p < 0.0001$), while evaluation of frozen selections on the holdout test set revealed a 12-point non-dominated set spanning both DOE and TPE configurations. These results demonstrate that classical DOE provides a mathematically rigorous, interpretable, and reproducible framework for tabular machine learning hyperparameter optimization.

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